



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

B.Sc. PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	INTRODUCTORY MATERIALS SCIENCE				
Type of Course	Vocational Minor (GROUP: TECHNIQUES IN MATERIALS PHYSICS)				
Semester	I				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Basics of Physics and Chemistry (Higher Secondary Level)				
Course Summary	Explore the diverse world of materials and their properties through experimentation and analysis.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Demonstrate a fundamental understanding of the different classes of materials (metals,	U	F	Instructor-created exams / Quiz

	ceramics, polymers, and composites) and their properties.			
CO2	Apply electrical and magnetic property concepts to analyze and design materials for various applications.	Ap	C	Instructor-created exams / Quiz
CO3	Explain the interaction of light with materials and its impact on optical properties.	An	C	Seminar Presentation / Group Tutorial Work
CO4	Relate thermal properties of materials to their behaviour in different temperature environments.	An	C	Instructor-created exams / Home Assignments
CO5	Develop practical skills in using laboratory equipment to measure and characterize material properties.	An	P	Practical Assignment / Observation of Practical Skills
CO6	Analyze and interpret experimental data to draw meaningful conclusions.	An	P	Practical Assignment / Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	Introduction to Materials and Properties		5	13
	1	Introduction to Materials science: Scope, applications, and interdisciplinary connections	1	

	2	Classification of materials: metals, ceramics, polymers, and composites.	2	
	3	Advanced materials: Semiconductors, Bio materials, Smart Materials, Nanomaterials, Modern Materials' Need	2	
	Sections 1.2 to 1.6, Book 1			
II	Electrical and Magnetic Properties		17	17
	4	Electrical properties: Ohm's law, electrical conductivity and resistivity	1	
	5	Ionic conduction	1	
	6	Electric conduction in terms of Band and atomic bonding models	2	
	7	Electron mobility	1	
	8	Semiconductors- intrinsic and extrinsic	2	
	9	Temperature dependence of carrier concentration, Hall Effect	3	
	10	Magnetic properties: diamagnetism and paramagnetism, ferromagnetism, antiferromagnetism and ferrimagnetism	3	
	11	Magnetic domains, magnetic hysteresis, soft and hard magnetic materials, Magnetic storage.	4	
	Sections 18.1 to 18.14, 20.1 to 20.12, Book 1			
III	Optical properties		14	25
	12	Electromagnetic radiation, light interaction with solids	2	
	13	Optical properties of metals- absorption and emission	3	
	14	Optical properties of nonmetals - refraction reflection absorption and transmission	3	
	15	Colour of transparent materials	2	
	16	Luminescence, photoconductivity	2	

	17	Light emitting diodes, lasers	2	
	Sections 21.1 to 21.13, Book 1			
IV	Thermal properties		9	15
	18	Heat capacity	2	
	19	Temperature dependence of heat capacity	2	
	20	Thermal expansion	2	
	21	Thermal conductivity	2	
	22	Thermal stress	1	
	Sections 19.1 to 19.5, Book 1			
V	PRACTICALS		30	
	1	Familiarization with laboratory equipment and safety protocols		
	2	Measuring resistivity of different materials using simple equipment		
	3	Measuring the refractive index of various samples		
	4	Measuring thermal conductivity using Lees disc method		

Books and References:

- 1) Materials Science and Engineering, An introduction by William D. Callister, Jr. David G. Rethwisch (Book 1)
- 2) Introduction to Materials Science for Engineers by James F. Shackelford
- 3) The Science and Engineering of Materials, by Donald R. Askeland and Pradeep P. Phulé

Mapping of COs with PSOs and POs :

	PS O1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	2	0	1	0	2	0	1	1	1	1	1	1	1
CO 2	2	1	2	0	1	0	1	1	1	1	2	1	1
CO 3	2	0	1	0	2	0	1	1	1	1	2	1	1

CO 4	2	0	1	0	1	0	1	1	1	1	1	1	1
CO 5	2	2	0	0	1	1	1	1	2	1	1	1	1
CO 6	2	2	1	0	0	1	1	1	1	1	2	1	1

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓			✓
CO 3	✓			✓
CO 4		✓		✓
CO 5		✓		✓
CO 6			✓	